

AMRIT PUHAN

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SUMMARY

Interested in how reliable collective judgments can be drawn from noisy, disagreeing people, and how that question carries over to evaluating and aligning AI systems with human preferences.

EDUCATION

Master of Science, Informatics (Data Science Concentration) 08/2022 – 08/2024
Pennsylvania State University, University Park, PA, USA GPA: 3.88/4.0
Thesis: *Recovering Ground Truth Rankings When the Majority Is Misinformed* (**Best MS Thesis Award in AI**).
Advisor: Dr. Hadi Hosseini

Bachelor of Technology, Computer Science and Engineering 07/2016 – 05/2020
National Institute of Technology, Rourkela, India GPA: 3.7/4.0
Advisor: Prof. Anup Nandy

PUBLICATIONS

- Hosseini, H., Mandal, D., **Puhan, A.*** SP-Rank: A Dataset for Ranked Preferences with Secondary Information. **KDD 2026**, Datasets and Benchmarks Track.
- Hosseini, H., Mandal, D., **Puhan, A.*** Surprisingly Popular Voting with Concentric Rank-Order Models. **The ACM Web Conference (WWW) 2025**.
- Hosseini, H., Mandal, D., **Puhan, A.*** The Surprising Effectiveness of SP Voting with Partial Preferences. **NeurIPS 2024**
- Hazra, S., Pisipati, M., **Puhan, A.**, Nandy, A., Scherer, R. Two Novel Methods for Multiple Kinect v2 Sensor Calibration. *CVIP 2021*, Springer.
- Pisipati, M.[†], **Puhan, A.**[†], Kumar, A., Semwal, V.B., Agrawal, H. A Dynamic Hand Gesture-Based Password Recognition System. *ICCIS 2019*, Springer.

*First author; authors listed alphabetically. [†]Equal contribution, co-first author.

RESEARCH EXPERIENCE

FAIR Lab, Pennsylvania State University, Graduate Research Assistant 01/2023 – 12/2024
Master's research advised by Dr. Hadi Hosseini, in collaboration with Dr. Debmalya Mandal (University of Warwick)

Ground-truth ranking recovery with Surprisingly Popular (SP) voting (MS Thesis)

- Designed scalable SP voting algorithms that recover ground-truth rankings from partial preferences when experts are a minority. The best variant (Partial-SP) reaches a Kendall-Tau correlation of about **0.8** with the true ranking, against about **0.2** for Copeland on votes alone. Proved that sample complexity grows with $k!$ for subsets of size k rather than $m!$ for all alternatives. (*NeurIPS 2024*)
- Ran a large-scale crowdsourcing study with **432** participants on Amazon MTurk across 3 domains (geography, movies, paintings) of **36** alternatives each.
- Modeled voter behavior as a concentric mixture of Mallows models with experts and non-experts, with parameters inferred by Bayesian MCMC (No-U-Turn Sampler). Simulated data closely reproduced real MTurk voting behavior.
- Generalized the analysis to concentric mixtures of Mallows and Plackett-Luce models with multiple expertise groups (expert, intermediate, non-expert identified in real data), deriving conditions and sample complexities for SP voting to recover the ground truth with high probability. (*WWW 2025*)
- Released **SP-Rank**, the first large-scale public dataset pairing personal votes with predictions of others' votes. It holds **12,384** datapoints from **1,152** participants across 9 elicitation formats, released open-source with benchmarks for rank recovery and voter modeling. (*KDD 2026*)

LLM fine-tuning and preference learning

- Ongoing work: exploring rank-aggregation methods for preference-based LLM fine-tuning (RLHF/RLAIF), with early experiments on Llama 3, Mistral-v3 and Gemma.
- Fine-tuned LLMs for multiple-choice QA (SWAG, ARC-Challenge) with LoRA, quantization and PEFT. Ranked **6th** on the A12 SWAG leaderboard with **90.98%** accuracy (2024).

Intelligent Computing and Computer Vision Group, NIT Rourkela, Undergraduate Researcher 08/2019 – 05/2020
Bachelor's thesis advised by Prof. Anup Nandy

- Proposed two calibration methods for an 8-sensor Microsoft Kinect v2 gait-capture setup, reaching a mean re-projection error of **0.42 pixels**, a **76.27%** reduction over existing methods. (*CVIP 2021*)
- Predicted cognitive state from calibrated gait features with ML models ($R^2 = 0.9556$), suggesting gait is a strong indicator of cognitive state.
- Co-developed a hand-gesture password recognition system that runs on an ordinary laptop webcam, reaching **96.16%** test accuracy, as co-first author. (*ICCIS 2019*)

Deep Learning Course Project, Penn State

01/2023 – 05/2023

- Built a two-stage breast cancer detector from mammograms that combines attention-based deep models with XGBoost over patient metadata, reaching an AUC-ROC of **0.81**.

TEACHING EXPERIENCE

Pennsylvania State University, Instructional Assistant, IST 261 (Java Application Development) 01/2023 – 05/2023

- Coached a 20+ student undergraduate class in object-oriented design, MVC architecture and debugging through office hours and one-on-one assignment help.
- Delivered a lecture on assignment design, graded with written feedback, and relayed anonymous student concerns to the instructor.

ACADEMIC SERVICE

Reviewer: The ACM Web Conference (WWW) 2025, Main Track; KDD 2026, Datasets and Benchmarks Track.

WORK EXPERIENCE

Amazon Web Services (AWS), Software Development Engineer II, Applied AI Solutions 12/2024 – Present

- Built a natural-language video search engine for autonomous-driving data curation that interprets free-text queries, searches video and its structured data, and ranks results for model training.
- Served as the team's primary science contact for time-series anomaly detection in predictive maintenance of industrial equipment, evaluating supervised and unsupervised statistical and deep-learning models.
- Contributed to an IoT service connecting vehicle fleets to the cloud for over-the-air software updates and data transfer.

Pennsylvania State University, Graduate Assistant, Organizational Effectiveness 08/2023 – 08/2024

- Built Power BI dashboards that gave Penn State World Campus insights into student enrollment.
- Created a reusable dashboard template that cut development time by **60%**, and improved tracked enrollment KPIs by **25%**.

SAP Labs India, Associate Software Developer (Bengaluru, India) 07/2020 – 07/2022

- Developed **five** front-end applications for SAP SuccessFactors, an enterprise B2B cloud product.
- Served as Quality Analyst for one year, overseeing quality standards for **six** projects and coordinating the quarterly testing event across SAP SuccessFactors **twice**.
- Served as CI/CD contact for **ten** teams, automating microservice deployments with Jenkins and shipping **40+** applications across **five** countries.

Schneider Electric India, Summer Research Intern (Bengaluru, India) 05/2019 – 07/2019

- Built a prototype IoT door-entry system with real-time text, audio and video communication, with web and mobile support across Android and iOS, reducing latency by **83.3%** versus existing standards.

SKILLS

Programming: Python, R, Stan, C/C++, Java, JavaScript, MATLAB, SQL

ML & Statistics: PyTorch, Hugging Face (Transformers, PEFT), XGBoost, CUDA, Bayesian inference / MCMC

Systems & Tools: AWS, Docker, Kubernetes, Jenkins, Git, React, MongoDB, Databricks, Power BI

HONORS AND AWARDS

- **Best MS Thesis Award in Artificial Intelligence**, Penn State (2025).
- **NeurIPS 2024 Scholar Award**. Covered conference registration and accommodation.
- **Fast Learner Award**, SAP Labs India (2021). For rapid skill acquisition and timely delivery.
- **Top 1%, JEE Main**. Among 1.4 million candidates (Ministry of Education, Govt. of India).
- **National Talent Search Examination Scholar**. National rank 245 of 1 million+; held through undergraduate studies.
- **Junior Mathematics Olympiad Scholarship**. State rank 8; awarded by the Chief Minister of Odisha.